FIRST REVIEW REPORT

AI Based Age Prediction and Age Progression System

Using Face Recognition & Deep Learning

Submitted in partial fulfillment of the requirements for

Master of Computer Applications (MCA)

IV Semester Project

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1. Title of the Project

AI Based Age Prediction and Age Progression System Using Face Recognition & Deep Learning

This project focuses on developing an intelligent web-based AI system that performs accurate age estimation from facial images and generates realistic age-progressed faces using advanced computer vision and deep learning techniques. The system integrates multi-face detection, CNN-based age prediction, and GAN-based age progression to provide comprehensive facial aging analysis and visualization.

2. Problem Definition / Abstract

Age estimation and facial aging simulation are crucial challenges in computer vision with significant applications across multiple domains including forensics, security, age-restricted access control, missing person identification, entertainment, and medical diagnosis. Despite advances in technology, several critical challenges persist in current age estimation and progression systems.

Manual age estimation methods remain subjective, time-consuming, and prone to significant human bias and error. Human observers can typically estimate age within a 5-10 year range, which is insufficient for applications requiring precise age determination. Furthermore, most existing tools are designed for single-face analysis and lack the capability to process group photos with multiple faces simultaneously, limiting their practical utility in real-world scenarios such as family photo analysis or crowd age demographics.

Age progression technology, which visualizes how a person might look at different ages, is currently dominated by expensive specialized software that requires professional expertise to operate. These commercial solutions are inaccessible to general users due to high costs, complex interfaces, and limited availability. Additionally, there is a notable absence of integrated, comprehensive web-based solutions that combine both age prediction and age progression capabilities in a single accessible platform with user-friendly authentication and result management features.

Existing systems often suffer from limitations including lack of multi-face processing capability, inability to preserve facial identity during age progression, computational inefficiency, limited accuracy across diverse age groups and demographics, and absence of proper result visualization and comparison features. These gaps create a significant barrier to widespread adoption and practical deployment of age estimation and progression technologies.

The motivation behind this project is to address these real-world challenges by developing a comprehensive AI-Based Age Prediction and Age Progression System that leverages state-of-the-art deep learning architectures. The proposed solution combines computer vision techniques with neural networks to create an intelligent, accessible, and accurate system for facial age analysis.

The proposed system is a full-stack web application utilizing Django REST Framework for the backend and Angular for the frontend, providing a modern and responsive user

interface. The core functionality integrates OpenCV for robust multi-face detection using Haar Cascade and DNN-based detectors, pre-trained Convolutional Neural Network (CNN) models including ResNet, MobileNet, and EfficientNet for efficient and accurate age estimation through transfer learning, and Generative Adversarial Network (GAN) models such as Age-cGAN and StarGAN for realistic age progression while preserving facial identity.

The system architecture follows a three-tier design with a presentation layer built using Angular and TypeScript, an API gateway layer implemented with Django REST Framework, and a business logic layer containing Python-based deep learning modules. MongoDB serves as the database for storing user data, uploaded images, analysis results, and progression outputs. The system provides secure user authentication, image upload functionality, real-time age prediction with bounding box visualization, age progression with side-by-side comparison views, and comprehensive result management including download capabilities.

By integrating multiple advanced techniques and providing an accessible web-based interface, this system aims to democratize age prediction and progression technologies, making them available to a broader audience including law enforcement agencies, healthcare providers, entertainment professionals, researchers, and general users interested in facial aging analysis. The solution emphasizes accuracy, efficiency, user experience, and practical applicability while maintaining strong academic foundations through comprehensive literature review and validation on benchmark datasets including UTKFace, IMDB-WIKI, MORPH, and CACD.

3. Related Work / Existing System

A comprehensive review of recent literature on age estimation and progression techniques reveals significant advances in deep learning approaches from 2021 to 2025. This section summarizes key research papers, methodologies, datasets, and performance outcomes that inform and motivate the proposed system.

Recent Research (2024-2025):

Deep Learning-Based Age Estimation and Gender Classification (2025) - This arXiv preprint demonstrates multi-task CNN architectures with shared feature extractors for simultaneous age and gender prediction using UTKFace and IMDB-WIKI datasets. The research shows that multi-task learning significantly outperforms single-task models in both accuracy and computational efficiency.

Real Time Face Aging Progression Using GANs (2025) - Published in IJCRT, this work implements real-time GAN variants including Progressive GAN, StarGAN, and StyleGAN with attention mechanisms on MORPH and CACD datasets. The system achieves high-resolution aged faces with low latency suitable for real-time applications, demonstrating substantial improvements in visual quality compared to earlier GAN architectures.

Face Analysis Technology Evaluation by NIST (2024) - This comprehensive NIST Report 8525 provides independent evaluation of commercial age estimation and verification systems using large-scale operational facial images. The benchmark

establishes accuracy baselines and error rates across different demographics, revealing that performance varies significantly by age, gender, and ethnicity.

Age Prediction Using Deep Learning Architecture by Dzenis et al. (2024) - Published in Applied Computer Systems, this research evaluates various deep CNN architectures including ResNet, MobileNet, and EfficientNet through transfer learning on MORPH-2, CACD, and AFAD datasets. Results demonstrate that EfficientNet models achieve the lowest Mean Absolute Error (MAE) and highest accuracy, significantly outperforming baseline approaches.

Clinical and Explainable AI Research (2023):

Exploring AI from Clinical Perspective by Li et al. (2023) - This clinical study compares two CNN-based facial age predictors trained on large-scale Chinese cosmetic patient databases. The models approximate biological and clinical age parameters, supporting medical decision-making in cosmetic and dermatological applications.

Face Aging by Explainable Conditional Adversarial Networks by Wu et al. (2023) - Published in Scientific Reports, this research introduces an explainable Conditional Adversarial Autoencoder (CAAE) based GAN for age progression and regression using CACD and MORPH datasets. The approach generates realistic age transformations with interpretable aging effects while maintaining strong identity preservation.

GAN-Based Age Progression Research (2022):

Comparative Analysis of CycleGAN and AttentionGAN by Wang et al. (2022) - This comparative study evaluates CycleGAN versus AttentionGAN architectures for facial age progression on CACD and MORPH datasets. Findings indicate that attention-based models produce more realistic texture changes and better preserve identity characteristics during age transformation.

Age Estimation Using CNN by Al-Quraishi et al. (2022) - This University of Baghdad research presents a CNN-based age estimation pipeline with supervised learning on IMDB-WIKI dataset, achieving high prediction accuracy with relatively simple CNN architectures, demonstrating the effectiveness of deep learning for age estimation tasks.

Apparent Age Prediction Survey by Fu et al. (2022) - This comprehensive open-access survey reviews CNN architectures, transfer learning approaches, ensemble methods, and multi-task strategies on APPA-REAL and ChaLearn LAP datasets. The structured comparison identifies optimal MAE values and highlights persistent challenges including label noise and demographic bias.

Foundational Research (2021):

Survey on Deep Learning Face Age Estimation by Shukri et al. (2021) - Published in IJACSA, this survey examines CNN, regression, ranking, and multi-task approaches across MORPH, FG-NET, IMDB-WIKI, and UTKFace datasets. The comprehensive analysis summarizes MAE and accuracy metrics while identifying critical issues including ethnicity bias and data imbalance that affect model generalization.

Prediction of Face Age Progression with GANs by Abdollahi et al. (2021) - This Healthcare (MDPI) publication introduces GAN-based face age progression models that capture texture and structural changes on MORPH and CACD datasets. The system generates realistic age-progressed faces with clear aging indicators while maintaining good identity preservation, establishing foundational techniques for age transformation.

Key Limitations of Existing Systems:

- Most systems focus exclusively on either age estimation OR age progression, lacking integrated solutions
- Limited multi-face detection and analysis capabilities for group photographs
- Commercial solutions are expensive, require specialized expertise, and lack accessibility
- Absence of comprehensive web-based platforms with user authentication and result management
- Performance degradation across diverse demographics, particularly for underrepresented groups
- Insufficient identity preservation during age progression in many GAN-based approaches

The proposed system addresses these limitations by providing a unified platform that combines state-of-the-art age prediction and progression capabilities with multi-face processing, accessible web interface, secure authentication, and comprehensive result visualization. The integration of transfer learning with pre-trained CNN models and advanced GAN architectures ensures both accuracy and efficiency while maintaining computational feasibility for real-world deployment.

4. Objectives and Scope of the Proposed Work

Objectives:

1. To develop an AI-based system for accurate age prediction from facial images using state-of-the-art deep learning techniques including Convolutional Neural Networks (CNNs) with transfer learning approaches.
2. To implement robust multi-face detection capability in group images with high precision using OpenCV's Haar Cascade and DNN-based face detection algorithms to enable simultaneous age analysis of multiple individuals.
3. To create a GAN-based age progression module using pre-trained models (Age-cGAN/StarGAN) that generates realistic future facial appearances while preserving individual identity and facial characteristics.
4. To build a comprehensive full-stack web application with secure user authentication, responsive interface, and intuitive user experience using modern frameworks (Angular and Django).
5. To integrate pre-trained CNN models (ResNet, MobileNet, EfficientNet) for efficient age estimation with minimal training overhead through transfer learning, achieving low Mean Absolute Error (MAE) on benchmark datasets.
6. To design and implement RESTful APIs for seamless frontend-backend communication, enabling efficient data exchange, request handling, and response processing between client and server components.
7. To evaluate system performance using standard metrics including Mean Absolute Error (MAE), accuracy, precision, and processing time on benchmark datasets (UTKFace, IMDB-WIKI, MORPH) to validate effectiveness and reliability.

Scope:

8. Development of a comprehensive web-based age prediction and progression system using Django REST Framework for backend API services and Angular framework with TypeScript for frontend user interface.
9. Implementation of face detection using OpenCV library with multiple detection algorithms including Haar Cascade classifiers for rapid detection and DNN-based detectors for improved accuracy in challenging conditions.
10. Age estimation module using transfer learning approach with pre-trained CNN models including ResNet (Residual Networks), MobileNet (lightweight architecture), and EfficientNet (compound scaling) fine-tuned on age-specific datasets.
11. Age progression implementation using pre-trained Generative Adversarial Network models (Age-cGAN for conditional age transformation and StarGAN for multi-domain age progression) with strong identity preservation mechanisms.
12. Complete user management system including secure registration and authentication, image upload with format validation and size restrictions, real-time result visualization with bounding boxes and age overlays, and comprehensive result history management.
13. MongoDB NoSQL database implementation for efficient storage and retrieval of user profiles, uploaded images with metadata, age prediction results with detected face coordinates, and age-progressed images with comparison data.

14. Comprehensive testing and validation on industry-standard benchmark datasets including UTKFace (20,000+ diverse faces), IMDB-WIKI (500,000+ celebrity images), MORPH (55,000+ longitudinal images), and CACD (160,000+ cross-age images) to ensure robustness and generalization.

5. Block Diagram / System Architecture

The system follows a modern three-tier client-server architecture designed for scalability, maintainability, and performance. The architecture separates concerns into distinct layers: presentation layer, API gateway layer, and business logic layer, each with specific responsibilities and technologies.

Architectural Components:

Layer 1: Presentation Layer (Frontend)

- Framework: Angular 15+ with TypeScript for type-safe development
- Technologies: HTML5, CSS3, TypeScript, RxJS for reactive programming
- Features: Responsive UI design, user authentication interface, image upload component, real-time result visualization, comparison view for age progression
- Communication: HTTP client for RESTful API calls, Observable patterns for asynchronous operations

Layer 2: API Gateway Layer (Backend Server)

- Framework: Django 4.x with Django REST Framework (DRF) for robust API development
- Responsibilities: RESTful endpoint management, request/response handling, authentication and authorization, input validation and sanitization
- Security: Token-based authentication (JWT), CORS configuration, request rate limiting
- API Endpoints: /api/auth/login, /api/auth/register, /api/upload, /api/predict-age, /api/progress-age, /api/results

Layer 3: Business Logic Layer (Core Processing)

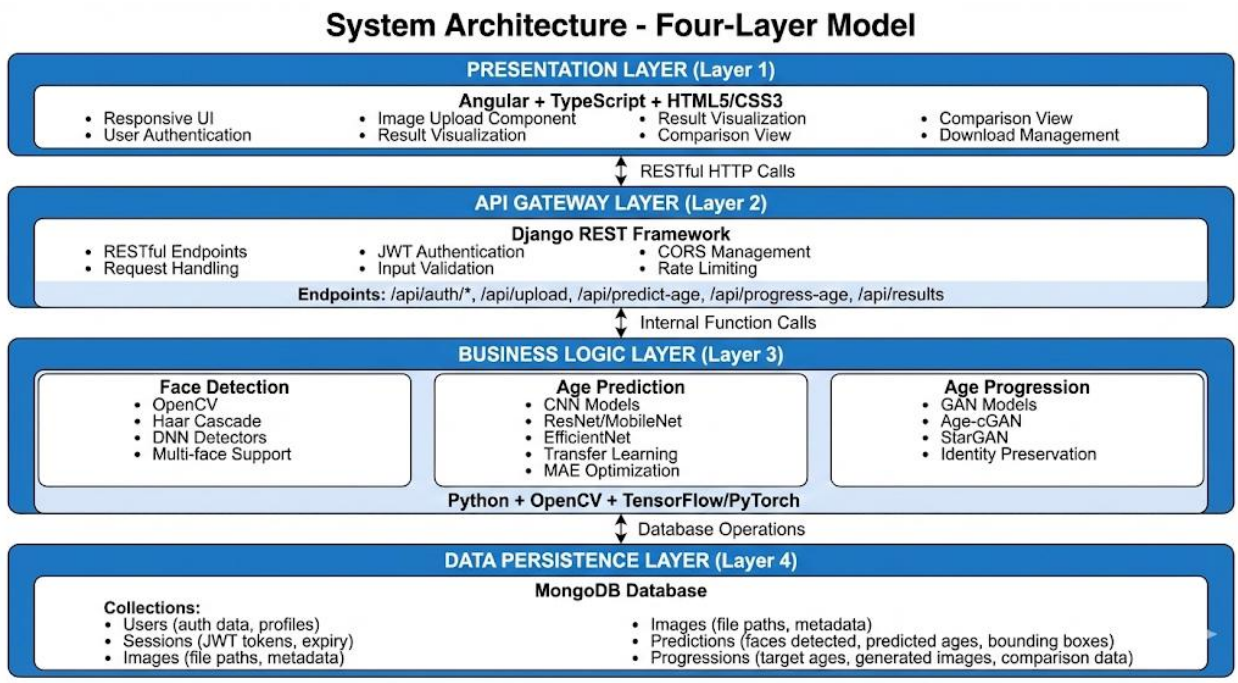
- Language: Python 3.8+ for deep learning model integration
- Libraries: OpenCV for computer vision operations, TensorFlow/PyTorch for deep learning, NumPy for numerical computations
- Modules: Face Detection Module (Haar Cascade + DNN detectors), Age Prediction Module (CNN with transfer learning), Age Progression Module (GAN-based generation)
- Processing Pipeline: Image preprocessing, face detection and extraction, age prediction inference, age progression generation, result formatting

Layer 4: Data Persistence Layer (Database)

- Database: MongoDB 5.0+ (NoSQL document database) for flexible schema and scalability
- Collections: Users (authentication data), Sessions (active sessions), Images (uploaded files metadata), Predictions (age estimation results), Progressions (age-progressed outputs)
- Features: Document-based storage, indexed queries for fast retrieval, GridFS for large file storage, replication for data redundancy
- ODM: PyMongo for Python-MongoDB integration, Mongoengine for object-document mapping

System Architecture Diagram:

The following diagram illustrates the complete system architecture showing the interaction between all layers:



Data Flow:

15. User interacts with Angular frontend (login, register, upload image)
16. Frontend sends HTTP requests to Django REST API endpoints
17. API Gateway validates request, authenticates user via JWT token
18. Business Logic Layer processes image (face detection → age prediction/progression)
19. Results stored in MongoDB with user references
20. API returns JSON response with results/file paths to frontend
21. Frontend renders results with visualizations and provides download options

6. Algorithm / Technique Proposed or Solution Methodology

The system implements two primary pipelines: Age Prediction Pipeline and Age Progression Pipeline. Each pipeline consists of multiple stages with specific algorithms and techniques designed for optimal performance and accuracy.

6.1 Age Prediction Pipeline

Stage 1: Face Detection and Preprocessing

Input: Raw facial image (single person or group photo)

Algorithm:

22. Load image using OpenCV (cv2.imread())
23. Convert to grayscale for Haar Cascade detection
24. Apply Haar Cascade Classifier (haarcascade_frontalface_default.xml) for rapid face detection
25. If low confidence or no faces detected, use DNN-based detector (OpenCV DNN module with Caffe model)
26. For each detected face: Extract bounding box coordinates (x, y, width, height)
27. Crop face region with padding (10-15% margin for context)
28. Resize to standard input size (224×224 pixels for CNN compatibility)
29. Normalize pixel values to [0, 1] range (divide by 255)

Output: List of preprocessed face images with coordinates

Stage 2: CNN-Based Age Estimation

Model Architecture: Transfer Learning with Pre-trained CNN

Primary Model: EfficientNet-B0 (chosen based on literature review showing best MAE)

Algorithm:

30. Load pre-trained EfficientNet-B0 model (trained on ImageNet)
31. Remove original classification head
32. Add custom regression head: GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.3) → Dense(1, Linear)
33. Fine-tune on UTKFace dataset: Freeze base layers → Train custom head (10 epochs) → Unfreeze top layers → Full training (20 epochs)
34. Loss function: Mean Absolute Error (MAE)
35. Optimizer: Adam with learning rate 0.0001
36. For inference: Pass preprocessed face through model → Output predicted age (regression value)
37. Round to nearest integer for display

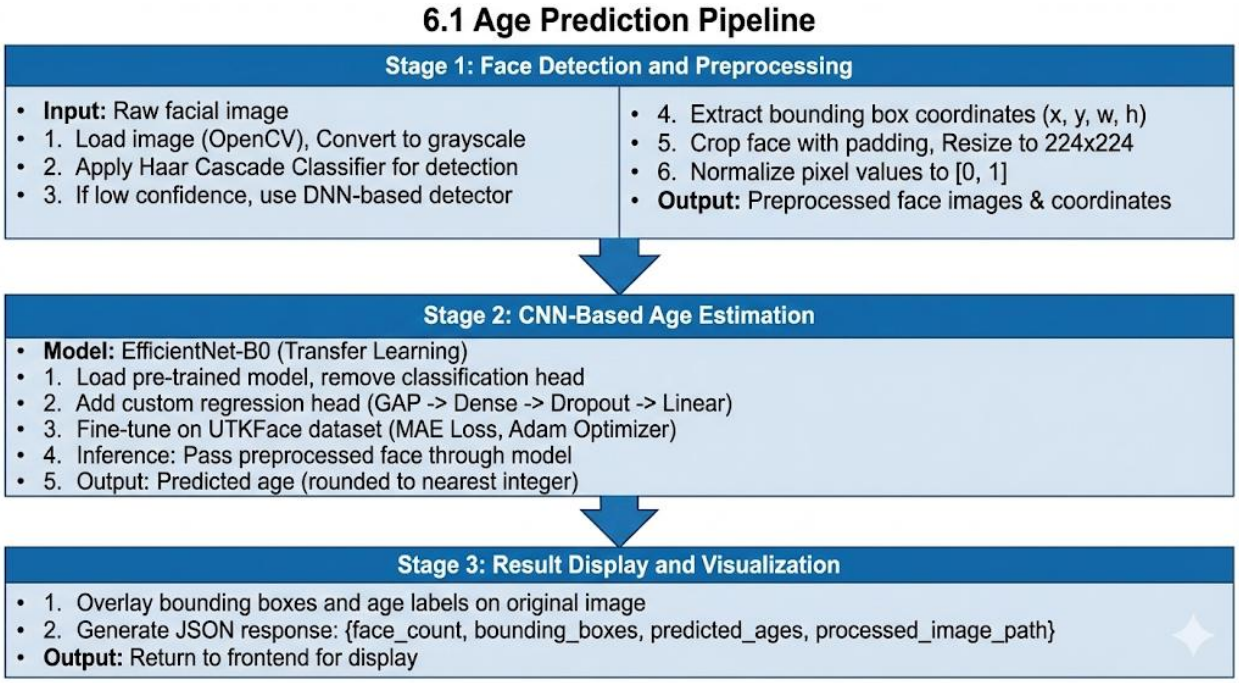
Output: Predicted age for each detected face

Stage 3: Result Display and Visualization

Algorithm:

38. Overlay bounding boxes on original image using OpenCV rectangle drawing
39. Add age labels above each bounding box (putText with white background for visibility)

40. Generate JSON response with: face_count, bounding_boxes[], predicted_ages[], processed_image_path
41. Return to frontend for display



6.2 Age Progression Pipeline

Stage 1: Face Detection (Same as Age Prediction Pipeline Stage 1)

Stage 2: Target Age Input

User specifies desired target age (e.g., 25, 40, 60, 80)

System validates: target_age >= current_estimated_age (only forward progression)

Stage 3: GAN-Based Age Progression

Model Architecture: Conditional Generative Adversarial Network (Age-cGAN) or StarGAN

Algorithm:

42. Load pre-trained Age-cGAN model (trained on MORPH/CACD datasets)
43. Generator Network: Encoder (downsample face) → Age Condition Vector (one-hot encoding of target age group) → Decoder (upsample with aging transformations)
44. Identity Preservation: Skip connections between encoder and decoder maintain facial features
45. Input: Preprocessed face + target age group label
46. Forward pass through generator: output = Generator(face, target_age_label)
47. Apply aging effects: wrinkles, skin texture changes, facial structure modifications
48. Post-process: Denormalize, convert to image format, resize to original dimensions

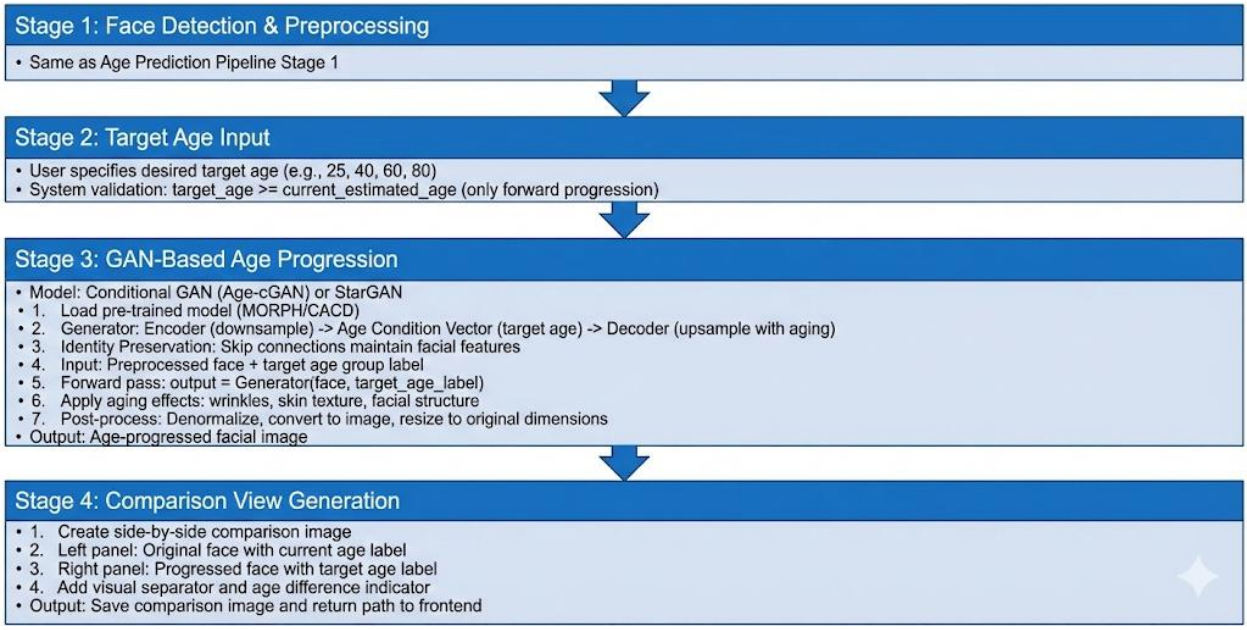
Output: Age-progressed facial image

Stage 4: Comparison View Generation

Algorithm:

49. Create side-by-side comparison image
50. Left panel: Original face with current age label
51. Right panel: Progressed face with target age label
52. Add visual separator and age difference indicator
53. Save comparison image and return path to frontend

6.2 Age Progression Pipeline



6.3 Optimization Techniques

- Model Quantization: Reduce model size and inference time using TensorFlow Lite or ONNX
- Batch Processing: Process multiple faces simultaneously using GPU acceleration
- Caching: Store frequently accessed model weights in memory to reduce loading time
- Asynchronous Processing: Use Celery task queue for long-running age progression tasks
- Image Compression: Optimize uploaded images before processing to reduce memory usage

7. Analysis and Design

7.1 Hardware and Software Requirements

Hardware Requirements:

Development Environment:

- Processor: Intel Core i5 or higher (8th generation+)
- RAM: Minimum 8 GB (16 GB recommended for model training)
- Storage: 50 GB free space (for datasets, models, and development tools)
- GPU: NVIDIA GPU with CUDA support (optional but recommended for faster training)

Production Environment:

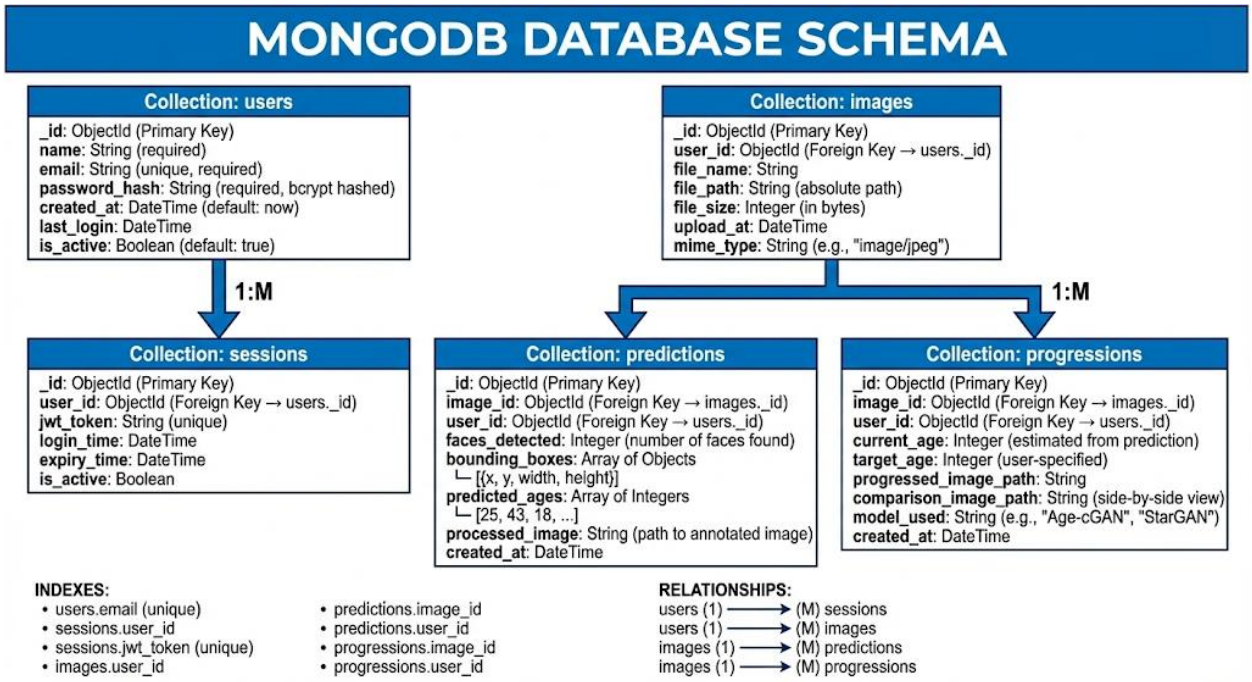
- Cloud Platform: AWS EC2, Google Cloud Platform, or Microsoft Azure
- Instance Type: 4 vCPUs, 16 GB RAM
- Storage: 100 GB SSD for application and user data

Software Requirements:

- Operating System: Windows 10/11, macOS Monterey+, or Linux (Ubuntu 20.04+)
- Backend: Python 3.8+, Django 4.x, Django REST Framework
- Frontend: Node.js 16+, Angular CLI 15+, TypeScript 4.8+
- Database: MongoDB 5.0+
- Deep Learning: TensorFlow 2.10+ or PyTorch 1.12+, OpenCV 4.6+
- Version Control: Git 2.30+
- Web Browser: Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- IDE: Visual Studio Code, PyCharm Professional, or IntelliJ IDEA

7.2 Database Table Design

The system uses MongoDB as the NoSQL database. Below are the main collections and their schema designs:



8. Work Plan

The project work is systematically divided into five major phases spanning requirement analysis, system design, module implementation, integration, testing, and final deployment. Each phase has specific deliverables and milestones to ensure timely completion and quality assurance.

9. Work Done So Far

- Comprehensive literature survey and requirement analysis completed
- Problem definition and scope finalization with clear objectives
- Complete system architecture design including three-tier architecture
- Module design and component specification for all layers
- Technology stack evaluation and final selection (Django, Angular, MongoDB)
- Database schema design with complete entity-relationship modeling
- API endpoint specification and RESTful architecture documentation
- UI/UX wireframes and mockup design for all user interfaces

10. Work To Be Done

- Angular frontend development and responsive UI implementation
- Django backend development with REST API endpoint creation
- Face detection module integration using OpenCV (Haar Cascade + DNN)
- CNN model integration for age prediction with transfer learning
- GAN model integration for age progression with identity preservation
- MongoDB database implementation and connection setup
- Frontend-backend API integration and comprehensive testing
- User acceptance testing, bug fixes, and final documentation preparation

11. Time Schedule

The project timeline is structured across 12 weeks with clearly defined milestones:

Phase	Activity	Duration
Phase 1	Requirement Analysis and System Design	Week 1 (Month 1)
Phase 2	Frontend Development (Angular + UI Components)	Week 2 (Month 1)
Phase 3	Backend Development (Django APIs + Database)	Week 3-4 (Month 1)
Phase 4	AI Model Integration (Face Detection, Age Prediction, Age Progression)	Week 5-6 (Month 2)
Phase 5	Integration, Testing, and Deployment	Week 7 (Month 2)
Phase 6	Final Documentation and Project Report	Week 8 (Month 2)

	Preparation	

12. References

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